

WHITMAN 5 ENE, NE, USA

WMO: 744360

Lat: 42.0680N

Lon: 101.4450W

Elev: 1140

StdP: 88.36

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94079

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.7	-18.1	-23.9	0.5	-20.2	-21.4	0.6	-17.7	14.8	-10.1	13.6	-8.6	4.4	N/A	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	34.0	18.4	32.1	18.6	30.4	18.3	21.6	28.3	20.8	27.4	20.1	26.9	5.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	16.7	23.9	18.8	15.7	23.5	18.0	14.8	22.9	68.8	28.4	65.8	27.5	62.8	26.9	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.4	9.9	DB	-24.7	37.0	1.9	2.2	-26.1	38.6	-27.2	39.9	-28.3	41.1	-29.7	42.7	(4)
(5)			WB	-24.5	22.8	2.0	0.8	-25.9	23.4	-27.0	23.9	-28.2	24.3	-29.6	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-3.1	-3.1	3.5	7.5	13.0	19.5	22.8	21.5	17.1	9.2	3.3	-3.2		
	DBStd	10.87	6.70	7.24	6.56	5.76	5.24	3.77	3.21	3.25	4.68	5.60	6.37	6.78		
	HDD10.0	1843	408	369	217	110	31	0	0	0	5	82	214	408		
	HDD18.3	3754	666	601	460	325	176	30	4	11	77	286	452	666		
	CDD10.0	1500	1	2	15	36	124	284	398	355	217	56	12	0		
	CDD18.3	370	0	0	0	1	12	64	144	108	39	2	0	0		
	CDH23.3	3953	0	0	8	32	172	688	1495	1036	470	50	3	0		
	CDH26.7	1559	0	0	0	6	44	244	666	414	174	11	0	0		
(14) Wind	WSAvg	4.8	4.8	4.9	5.1	5.8	5.3	4.8	4.2	4.1	4.4	4.8	4.8	4.7		
(15) Precipitation	PrecAvg	525	10	12	28	62	73	99	74	56	44	44	17	9		
	PrecMax	694	22	31	49	113	159	148	179	95	90	99	51	18		
	PrecMin	298	1	0	5	8	22	16	19	22	15	1	1	0		
	PrecStd	94	6	8	12	29	36	36	39	20	23	25	13	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	17.7	24.4	27.4	30.7	34.9	37.6	35.5	34.0	28.1	23.6	14.5		
		MCWB	5.9	7.2	10.2	11.4	16.0	18.3	18.7	18.0	17.0	13.2	10.0	5.8		
	2%	DB	11.4	13.3	20.3	23.3	27.8	31.9	34.6	33.0	31.0	24.3	18.6	10.8		
		MCWB	4.6	5.4	8.5	11.5	15.2	18.7	18.8	18.2	16.8	12.2	8.1	4.1		
	5%	DB	8.3	9.8	17.1	20.4	25.0	29.5	32.5	30.9	28.6	21.3	15.5	8.3		
		MCWB	2.9	3.6	7.4	10.0	14.5	18.2	19.5	18.1	16.6	11.1	6.8	2.7		
	10%	DB	5.8	6.8	13.9	17.3	22.4	27.3	30.5	29.0	25.9	18.3	12.6	5.8		
		MCWB	1.7	2.1	5.8	8.8	13.1	17.6	19.3	18.2	15.3	9.9	5.7	1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.1	11.9	13.3	18.2	21.9	22.8	22.0	20.5	15.1	10.7	6.7		
		MCDB	13.3	17.3	20.9	22.1	25.2	28.4	30.0	29.0	28.5	23.7	18.9	12.4		
	2%	WB	4.3	5.8	9.7	12.0	17.0	20.8	21.9	21.0	19.1	13.4	8.6	4.7		
		MCDB	9.9	13.3	18.8	20.2	24.2	27.0	28.5	27.5	27.0	21.0	17.3	10.6		
	5%	WB	2.9	4.1	8.3	10.7	15.6	19.8	21.2	20.2	18.0	12.0	7.0	2.9		
		MCDB	7.2	9.3	16.8	19.1	21.9	27.0	28.0	26.5	25.3	18.7	14.7	7.8		
	10%	WB	1.7	2.5	6.8	9.5	14.2	18.8	20.4	19.5	16.9	10.6	5.6	1.6		
		MCDB	5.5	7.0	14.3	16.2	20.3	25.5	27.3	25.6	23.3	17.6	11.8	5.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	10.7	12.1	12.0	11.8	12.4	12.8	12.6	13.2	12.1	11.7	9.8		
		MCDBR	13.7	14.3	17.1	17.2	15.8	15.4	15.6	15.7	16.8	16.9	15.8	12.5		
		MCWBR	7.9	8.2	8.4	7.7	6.6	5.9	4.8	5.3	5.9	7.7	7.7	7.7		
		MCDBR	12.3	14.5	16.8	15.7	13.9	13.9	13.4	13.0	14.4	15.4	15.0	12.5		
	5% WB	MCWBR	7.9	8.7	8.5	7.6	6.4	6.5	5.6	5.2	6.4	7.6	7.6	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.259	0.282	0.312	0.334	0.351	0.373	0.367	0.328	0.289	0.261	0.253		
	taud		2.556	2.533	2.504	2.423	2.443	2.422	2.400	2.398	2.485	2.562	2.562	2.531		
	Ebn at Noon		926	968	974	959	939	920	895	891	909	916	903	890		
	Edh at Noon		67	80	94	110	111	114	116	112	95	78	66	63		
(44) All-Sky Solar Radiation	RadAvg		2.24	3.14	4.27	5.28	6.21	7.11	7.10	6.10	4.95	3.44	2.44	1.91		
	RadStd		0.15	0.17	0.32	0.28	0.45	0.46	0.23	0.26	0.24	0.31	0.14	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page